代数拓扑 APS 面试准备

Croom《Basic Concepts of Algebraic Topology》第 1–5 章

Algebraic Topology: Bilingual Review

课程概述 | Course overview

中文口述稿。我按教材顺序学习了几何复形、单纯同调、单纯逼近、基本群和覆盖空间。主线是先把空间剖分成顶点、边、三角形等简单部分，再用边界算子判断闭合链能否被高一维的面填充；随后研究连续映射怎样由有限的单纯映射近似；最后用闭路的拼接定义基本群，并借助覆盖空间把闭路的同伦类转化为提升路径的终点信息。例如，未填充的三角形边框有一个独立的一维洞，故 $H_1 \cong \mathbb{Z}$；填入三角面后该圈成为边界，$H_1 = 0$。这说明同调是可计算的代数不变量，而基本群保留更精细的闭路拼接信息。

English. Following the textbook, I studied geometric complexes, simplicial homology, simplicial approximation, the fundamental group, and covering spaces. The main idea is to decompose a space into vertices, edges, triangles, and higher-dimensional pieces, then use boundary maps to decide which cycles can be filled by higher-dimensional faces. Simplicial approximation turns continuous maps into finite combinatorial maps. The fundamental group records loops and their concatenation, while covering spaces encode loop classes through the endpoints of lifted paths. For example, an unfilled triangular frame has one independent one-dimensional hole, so $H_1 \cong \mathbb{Z}$. Filling its triangular face makes that loop a boundary and gives $H_1 = 0$. Homology is a computable algebraic invariant, while the fundamental group retains finer information about loop concatenation.

具体应用联系。在拓扑数据分析中，可以把传感器或三维扫描得到的点云按距离阈值连接成边，并在三点两两相连时加入三角形，得到随尺度变化的单纯复形。计算 H_0 可观察数据簇何时合并，计算 H_1 可观察环形空隙何时出现或被填掉。若一个洞只在极窄的尺度范围存在，它可能来自测量噪声；持续跨越多个尺度的特征通常更值得进一步检查。这个过程分析的是有限点云构成的模型，结论仍要结合采样密度和噪声判断。

English. In topological data analysis, a point cloud from sensors or a 3D scan can be turned into a simplicial complex at each distance scale: connect nearby points by edges and add a triangle when all three edges are present. The group H_0 tracks when data clusters merge, and H_1 tracks when loop-shaped gaps appear or are filled. A feature that exists only over a very narrow range of scales may be caused by measurement noise, while a feature persisting across scales deserves closer attention. The calculation concerns a finite model of the point cloud, so conclusions still depend on sampling density and noise.

1 第 1 章：几何复形与多面体 | Chapter 1: Geometric complexes and polyhedra

1.1 单形、面与仿射独立 | Simplices, faces, and affine independence

在欧氏空间中，k-单形是 $k+1$ 个仿射独立顶点的凸包。仿射独立表示顶点不落在一个维数更低的仿射子空间内。于是 0-单形是点，1-单形是闭线段，2-单形是实心三角形，3-单形是实心四面体。单形的面由它的任意一组顶点的凸包给出，空面通常也作为面处理。

English. A geometric k-simplex is the convex hull of $k+1$ affinely independent vertices. Affine independence means that the vertices do not lie in a lower-dimensional affine subspace. Thus a zero-simplex is a point, a one-simplex is a closed line segment, a two-simplex is a filled triangle, and a three-simplex is a solid tetrahedron. A face is the convex hull of any subset of the vertices; the empty face is usually included by convention.

例子 / Example. 顶点 $(0,0),(1,0),(0,1)$ 张成的二维单形是 $\{(x,y) : x \geq 0,\ y \geq 0,\ x+y \leq 1\}$。它的三条边是三个一维面；只保留这三条边时得到三角形边框，保留整个凸包时得到填充三角形。两者的顶点和边相同，但同调不同。

English. The vertices $(0,0)$, $(1,0)$, and $(0,1)$ span the simplex $\{(x,y) : x \geq 0, y \geq 0, x+y \leq 1\}$. Its three edges are one-dimensional faces. Keeping only those edges gives a triangular frame; keeping the whole convex hull gives a filled triangle. They have the same vertices and edges but different homology.

1.2 单纯复形、几何实现与三角剖分 | Complexes, realizations, and triangulations

单纯复形 K 是一族单形，并满足两个条件：每个单形的所有面也属于 K；任意两个单形的交为空，或是它们共同的面。它的几何实现 $|K|$ 是这些单形按规定粘合后形成的空间。若空间 X 同胚于某个 $|K|$，则 K 是 X 的一个三角剖分。三角剖分把连续几何对象转换成有限组合数据，但同一个空间可以有不同的三角剖分。

English. A simplicial complex K is a collection of simplices satisfying two conditions: every face of each simplex also belongs to K, and two simplices intersect only in a common face or not at all. Its geometric realization $|K|$ is the space obtained by gluing these simplices along their prescribed faces. If a space X is homeomorphic to $|K|$, then K is a triangulation of X. Triangulation converts a geometric space into finite combinatorial data, although the same space can have different triangulations.

例子 / Example. 三个顶点及其三条边组成、但不含二维单形的复形实现为圆周 S^1。加入由这三个顶点张成的二维单形后，几何实现变成闭圆盘。新增的只是一个面，却改变了边界闭路是否能被填充。

English. The complex consisting of three vertices and their three edges, but no two-simplex, realizes the circle S^1. Adding the two-simplex spanned by those vertices changes the realization into a closed disk. Only a face has been added, but whether the boundary loop can be filled has changed.

1.3 定向与边界符号 | Orientation and boundary signs

对单形顶点排序 $[v_0, \dots, v_k]$ 指定一个方向。交换两个顶点会反转方向，因此有 $[v_1, v_0] = -[v_0, v_1]$。计算链时，方向用正负号记录；相邻面共享的边界部分以相反方向出现，因而相消。这是后面边界公式中交替符号的几何来源。

English. An ordering $[v_0, \dots, v_k]$ of the vertices specifies an orientation. Swapping two vertices reverses the orientation, so $[v_1, v_0] = -[v_0, v_1]$. In chain calculations, orientation is recorded by signs. A shared face of adjacent simplices appears with opposite orientations and cancels; this is the geometric reason for the alternating signs in the boundary formula.

例子 / Example. 将三角形边界定向为 $v_0 \to v_1 \to v_2 \to v_0$。若把最后一条边写成 $[v_0, v_2]$，它的方向与沿边界行走的 $[v_2, v_0]$ 相反，所以应使用 $-[v_0, v_2]$。

English. Orient the triangle boundary as $v_0 \to v_1 \to v_2 \to v_0$. If the last edge is written as $[v_0, v_2]$, its direction is opposite to the boundary direction $[v_2, v_0]$, so it must appear as $-[v_0, v_2]$.

可能问题 / Possible question. 为什么单纯复形必须包含每个单形的所有面？

English. Why must a simplicial complex contain every face of each simplex?

回答要点 / Key answer. 边界算子要把一个单形送到它的面组成的链中。若这些面不在复形里，边界就不属于该复形的链群，后续的链复形和同调也无法定义。

English. The boundary map sends a simplex to a chain made of its faces. If those faces were absent, the boundary would not belong to the chain group of the complex, so the chain complex and its homology could not be defined.

2 第 2 章：单纯同调群 | Chapter 2: Simplicial homology groups

2.1 链群与边界算子 | Chain groups and boundary maps

选定系数 $\mathbb{Z}$ 后，$C_k(K;\mathbb{Z})$ 是由 K 的定向 k-单形生成的自由阿贝尔群；一个 k-链是有限个 k-单形的整数线性组合。反转单形方向等同于将相应生成元乘以 -1。边界算子 $\partial_k : C_k \to C_{k-1}$ 对一个有序单形定义如下，并按线性性扩张到所有链：

公式名称 / Formula name. 单纯边界公式（交替删点公式）/ the simplicial boundary formula, or the alternating face formula

$$\partial_k[v_0, \dots, v_k] = \sum_{j=0}^{k} (-1)^j [v_0, \dots, \widehat{v_j}, \dots, v_k].$$

公式英文读法 / How to read it. “The boundary of the simplex with vertices v zero through v k is the sum, from j equals zero to k, of minus one to the j times the face obtained by omitting v j.” The hat means that the marked vertex is omitted.

English. With integer coefficients, $C_k(K;\mathbb{Z})$ is the free abelian group generated by oriented k-simplices. A k-chain is a finite integer linear combination of these simplices. Reversing orientation multiplies a generator by -1. The boundary map is defined by the alternating face formula above and extended linearly to every chain.

例子 / Example. 对有向边，$\partial_1[v_0, v_1] = [v_1] - [v_0]$，即终点减起点。对三角形，$\partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$。中间的负号来自删去第二个顶点时的 $(-1)^1$。

English. For an oriented edge, $\partial_1[v_0, v_1] = [v_1] - [v_0]$, terminal endpoint minus initial endpoint. For a triangle, the boundary is the alternating sum shown above. The middle minus sign comes from the factor $(-1)^1$ when the second vertex is omitted.

2.2 边界的边界为零 | The boundary of a boundary is zero

对一个 k-单形连续取两次边界时，每个余维二的面会出现两次：一次先删去顶点 v_i 再删去 v_j，另一次按相反顺序删去。两项的符号相反，故逐项相消。于是得到**边界复形恒等式：**

公式名称 / Formula name. 边界的边界为零 / the boundary of a boundary is zero

$$\partial_{k-1} \circ \partial_k = 0.$$

公式英文读法 / How to read it. “The boundary in dimension k minus one, composed with the boundary in dimension k, is the zero map.”

English. Applying the boundary twice to a simplex gives each codimension-two face twice: once by omitting v_i and then v_j, and once in the reverse order. The two signs are opposite, so the terms cancel. This gives the identity above, often called “the boundary of a boundary is zero.”

这个恒等式保证每个边界都是圈：若 $b = \partial_{k+1} c$，则 $\partial_k b = \partial_k \partial_{k+1} c = 0$。因此 $\operatorname{im} \partial_{k+1} \subseteq \ker \partial_k$，后面才能把边界群作为圈群的子群取商。

例子 / Example. 有向三角形的边界是 $z = [v_0, v_1] + [v_1, v_2] + [v_2, v_0]$。再取一次边界得到 $(v_1 - v_0) + (v_2 - v_1) + (v_0 - v_2) = 0$，因为每个顶点一次以正号、一次以负号出现。

English. The oriented boundary of a triangle is $z = [v_0, v_1] + [v_1, v_2] + [v_2, v_0]$. Taking its boundary gives $(v_1 - v_0) + (v_2 - v_1) + (v_0 - v_2) = 0$, since each vertex occurs once with each sign.

2.3 圈、边界与同调群 | Cycles, boundaries, and homology

k-圈是边界为零的 k-链，组成 $Z_k = \ker \partial_k$；k-边界是某个 $(k+1)$-链的边界，组成 $B_k = \operatorname{im} \partial_{k+1}$。因为 $B_k \subseteq Z_k$，定义第 k 个同调群为圈群除以边界群：

公式名称 / Formula name. 第 k 个同调群的定义 / the definition of the k-th homology group

$$H_k(K; \mathbb{Z}) = Z_k / B_k = \ker \partial_k / \operatorname{im} \partial_{k+1}.$$

公式英文读法 / How to read it. “H k is the kernel of the boundary map in dimension k, modulo the image of the boundary map in dimension k plus one.”

English. A k-cycle is a k-chain with zero boundary; the cycle group is $Z_k = \ker \partial_k$. A k-boundary is the boundary of a $(k+1)$-chain; the boundary group is $B_k = \operatorname{im} \partial_{k+1}$. Since every boundary is a cycle, the k-th homology group is the quotient of cycles by boundaries.

为什么要取这个商？圈是闭合的候选“洞边界”，但若它恰好是某个面或高维链的边界，就能在空间内部填掉，不应算作非平凡洞。同调类 $[z]$ 把相差一个边界的圈视为等价；$[z] = 0$ 当且仅当 z 本身是边界。H_0 记录连通分支，H_1 记录不能由二维面填掉的一维圈，H_2 则记录不能由三维链填掉的二维闭合表面。

English. Why take this quotient? A cycle is a candidate boundary of a hole, but if it is itself the boundary of a face or a higher-dimensional chain, it can be filled inside the space and should not count as a nontrivial hole. Homology identifies cycles that differ by a boundary. A class $[z]$ is zero

exactly when z is a boundary. The group H_0 records connected components, H_1 records loops not filled by two-dimensional faces, and H_2 records closed surfaces not filled by three-dimensional chains.

2.4 完整计算：三角形边框与填充三角形 | Worked computation: a triangular frame and a filled triangle

先看未填充的三角形边框。令顶点为 v_0, v_1, v_2，边按环向定向为 e_{01}, e_{12}, e_{20}。链群为 $C_1 \cong \mathbb{Z}^3$、$C_0 \cong \mathbb{Z}^3$，而 $C_2 = 0$。按上述基底，边界矩阵为

公式名称 / Formula name. 三角形边框的一维边界矩阵 / the boundary matrix of the triangular frame

$$[\partial_1] = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}.$$

公式英文读法 / How to read it. “The boundary matrix has columns equal to the boundaries of e zero one, e one two, and e two zero, in that order.”

English. First consider the unfilled triangular frame. Let its vertices be v_0, v_1, v_2, and orient the edges cyclically as e_{01}, e_{12}, e_{20}. Then $C_1 \cong \mathbb{Z}^3$, $C_0 \cong \mathbb{Z}^3$, and $C_2 = 0$. The columns of the displayed matrix are the boundaries of the three oriented edges.

若 $c = ae_{01} + be_{12} + ce_{20}$，则 $\partial_1 c = 0$ 等价于 $a = b = c$，所以 Z_1 由 $e_{01} + e_{12} + e_{20}$ 生成。由于没有二维面，$B_1 = 0$，故 $H_1 \cong \mathbb{Z}$。矩阵秩为 2，图连通，所以 $H_0 \cong \mathbb{Z}$。这一个整数生成元表示沿边框绕行的次数；正整数和负整数分别表示两个方向的多次绕行。

English. For $c = ae_{01} + be_{12} + ce_{20}$, the equation $\partial_1 c = 0$ is equivalent to $a = b = c$. Thus Z_1 is generated by $e_{01} + e_{12} + e_{20}$. There is no two-simplex, so $B_1 = 0$ and $H_1 \cong \mathbb{Z}$. The matrix has rank two, and the graph is connected, so $H_0 \cong \mathbb{Z}$. The integer generator records the number of turns around the frame; positive and negative integers indicate repeated turns in opposite directions.

例子 / Example. 现在在同一边框中加入定向三角面 $t = [v_0, v_1, v_2]$。按边的环向定向，$\partial_2 t = e_{01} + e_{12} + e_{20}$，恰好是上面的圈生成元。因此 $B_1 = Z_1$，$H_1 = 0$。填充三角形本身没有二维洞，所以 $H_2 = 0$；空间仍连通，$H_0 = \mathbb{Z}$。两种复形的同调分别是边框：$H_0 = \mathbb{Z}, H_1 = \mathbb{Z}$；填充三角形：$H_0 = \mathbb{Z}, H_1 = H_2 = 0$。

English. Now add the oriented face $t = [v_0, v_1, v_2]$. With the cyclic edge orientations, its boundary is $e_{01} + e_{12} + e_{20}$, exactly the generator of the cycle group. Hence $B_1 = Z_1$ and $H_1 = 0$. The filled triangle has no two-dimensional hole, so $H_2 = 0$; it remains connected, so $H_0 = \mathbb{Z}$. Thus the frame has $H_0 = \mathbb{Z}, H_1 = \mathbb{Z}$, whereas the filled triangle has $H_0 = \mathbb{Z}$ and $H_1 = H_2 = 0$.

可能问题 / Possible question. 为什么三角形边框的 H_1 是 $\mathbb{Z}$，填充后却变成零？

English. Why is the first homology of the triangular frame $\mathbb{Z}$, but zero after filling it?

回答要点 / Key answer. 边框中的环是圈，但没有二维面能以它为边界，所以它代表非零同调类。填入三角面后，同一个环成为该面的边界，在商群 Z_1/B_1 中归零。

English. The loop in the frame is a cycle, but no two-dimensional face has it as boundary, so it represents a nonzero homology class. After adding the triangle, the same loop is a boundary and becomes zero in Z_1/B_1.

2.5 Betti 数、边界矩阵秩与 Euler–Poincaré 公式 | Betti numbers, matrix ranks, and the Euler–Poincare formula

若系数取域 F，第 k 个 Betti 数定义为 $\beta_k = \dim_F H_k(K;F)$，表示独立的第 k 维同调类数量。令 f_k 为 k-单形数，$r_k = \operatorname{rank} \partial_k$，则由秩-零度定理得到常用的边界矩阵计算式：

公式名称 / Formula name. Betti 数的秩公式 / the rank formula for Betti numbers

$$\beta_k = f_k - r_k - r_{k+1}.$$

公式英文读法 / How to read it. “The k-th Betti number equals the number of k-simplices minus the rank of the k-th boundary map minus the rank of the next boundary map.”

English. Over a field F, the k-th Betti number is $\beta_k = \dim_F H_k(K;F)$, the number of independent homology classes in dimension k. If f_k is the number of k-simplices and $r_k = \operatorname{rank} \partial_k$, the rank-nullity theorem gives the displayed rank formula.

对有限单纯复形，Euler 特征是单形数的交替和；Euler–Poincaré 公式说明它也等于 Betti 数的交替和：

公式名称 / Formula name. Euler 特征与 Euler–Poincaré 公式 / the Euler characteristic and the Euler–Poincare formula

$$\chi(K) = \sum_{k\geq 0} (-1)^k f_k = \sum_{k\geq 0} (-1)^k \beta_k.$$

公式英文读法 / How to read it. “The Euler characteristic is the alternating sum of the numbers of simplices, and it equals the alternating sum of the Betti numbers.”

English. For a finite simplicial complex, the Euler characteristic is the alternating sum of the numbers of simplices. The Euler–Poincare formula states that it also equals the alternating sum of the Betti numbers.

例子 / Example. 四面体表面是球面 S^2 的三角剖分，有 $f_0 = 4, f_1 = 6, f_2 = 4$，所以 $\chi = 4-6+4 = 2$。球面的 Betti 数为 $\beta_0 = 1, \beta_1 = 0, \beta_2 = 1$，交替和也是 $1 - 0 + 1 = 2$。Euler 特征是有用的快速检查，但不同空间可能有相同 Euler 特征，因此它不能单独判定空间同胚。

English. The boundary of a tetrahedron triangulates the sphere S^2. It has $f_0 = 4, f_1 = 6, f_2 = 4$, so $\chi = 4 - 6 + 4 = 2$. The Betti numbers of the sphere are $\beta_0 = 1, \beta_1 = 0, \beta_2 = 1$, whose alternating sum is also two. Euler characteristic is a useful check, but different spaces can have the same Euler characteristic, so it cannot by itself determine homeomorphism.

使用整数系数时，同调群可能还含有有限阶元素（挠元）；Betti 数只记录自由部分的秩。要报告完整的整数同调，必须说明挠部分，而不只是列出 Betti 数。

English. With integer coefficients, homology groups may also contain finite-order elements called torsion. Betti numbers record only the rank of the free part. To state the full integer homology, one must include any torsion, not just the Betti numbers.

3 第 3 章：单纯逼近 | Chapter 3: Simplicial approximation

3.1 单纯映射与诱导同调映射 | Simplicial maps and induced maps on homology

单纯映射 $\phi : K \to L$ 首先把顶点映到顶点，并要求每个单形的所有顶点像共同落在 L 的某个单形中。不同顶点可以映到同一顶点，故单形维数可能下降。它在线性单形上延拓为几何实现之间的连续映射，并诱导链映射 $\phi_\# : C_k(K) \to C_k(L)$。边界与映射相容，即 $\partial\phi_\# = \phi_\#\partial$，因此圈映到圈、边界映到边界，进而得到同调同态 $\phi_* : H_k(K) \to H_k(L)$。

English. A simplicial map $\phi : K \to L$ maps vertices to vertices and sends the vertices of each simplex into the vertices of a simplex of L. Distinct vertices may have the same image, so dimension can drop. Linear extension gives a continuous map between geometric realizations and a chain map $\phi_\# : C_k(K) \to C_k(L)$. It commutes with the boundary map, so it sends cycles to cycles and boundaries to boundaries and induces a homomorphism $\phi_* : H_k(K) \to H_k(L)$.

例子 / Example. 把三角形边框的三个顶点都映到单点复形的唯一顶点，是一个单纯映射。边框的生成圈被映成零链，因而诱导映射把 $H_1 \cong \mathbb{Z}$ 的生成元送到零。这个例子说明连续映射可以消去洞；同调群在映射下并非总保持不变，而是由映射诱导一个同态。

English. Map all three vertices of a triangular frame to the only vertex of a one-point complex. This is a simplicial map. The generating loop is mapped to the zero chain, so the induced map sends the generator of $H_1 \cong \mathbb{Z}$ to zero. A continuous map can destroy a hole: homology is not necessarily unchanged, but the map induces a homomorphism between homology groups.

3.2 重心细分与单纯逼近定理 | Barycentric subdivision and the simplicial approximation theorem

一般连续映射不一定把原剖分的顶点和单形映成目标中的单形，所以需要先把定义域剖分得更细。重心细分把每个单形切成由其各级面的重心组成的小单形；反复细分会使最大单形直径趋于零。顶点 v 的开星由所有含 v 的单形的相对内部组成。

English. A general continuous map need not send the vertices and simplices of the original triangulation to simplices in the target, so the domain may need a finer triangulation. Barycentric subdivision divides each simplex into smaller simplices whose vertices are barycenters of nested faces. Repeated subdivision makes the largest simplex diameter tend to zero. The open star of a vertex v is the union of the relative interiors of simplices containing v.

单纯逼近定理。若 K, L 是有限单纯复形，且 $f : |K| \to |L|$ 连续，则对 K 作足够多次重心细分后，可以找到单纯映射 ϕ，使几何实现 $|\phi|$ 与 f 同伦。常用的顶点选择条件是：对细分后每个顶点 v，选取目标顶点 w，使 $f(\mathrm{st}(v)) \subseteq \mathrm{st}(w)$。有限性给出紧致性，连续性与紧致性保证细分足够细时可以满足该条件。这里的结论是“同伦于”，通常不是逐点相等。

English. The simplicial approximation theorem says that if K, L are finite simplicial complexes and $f : |K| \to |L|$ is continuous, then after sufficiently many barycentric subdivisions of K, there is a simplicial map ϕ whose realization is homotopic to f. A common vertex-selection condition is to choose, for each vertex v of the subdivision, a target vertex w such that $f(\mathrm{st}(v)) \subseteq \mathrm{st}(w)$. Finiteness gives compactness, and continuity plus compactness ensures that sufficiently fine subdivisions can meet this condition. The conclusion is “homotopic to,” not usually pointwise equality.

同伦映射诱导相同的同调同态，所以单纯逼近把连续问题转成有限顶点映射和边界矩阵计算。特别地，同伦等价的空间有同构的同调群；这解释了为什么三角剖分的选择不会改变所研究空间的同调。

English. Homotopic maps induce the same homomorphism on homology. Simplicial approximation therefore converts a continuous problem into a finite vertex map and boundary-matrix calculation. In particular, homotopy-equivalent spaces have isomorphic homology groups, so changing a triangulation does not change the homology of the underlying space.

例子 / Example. 若 f 是圆周三角剖分上的恒等映射，顶点可映回自身，单纯逼近就是原映射。对更一般的连续映射，先细分圆周，再按开星条件选目标顶点；得到的顶点映射可写成有限矩阵，并计算它对 H_1 的作用。

English. For the identity map on a triangulated circle, each vertex can be sent to itself, and the simplicial approximation is the original map. For a general continuous map, first subdivide the circle and then choose target vertices using the open-star condition. The resulting vertex map is finite and can be used to compute the induced map on H_1.

可能问题 / Possible question. 单纯逼近为什么要先细分？

English. Why may subdivision be needed before simplicial approximation?

回答要点 / Key answer. 原来的单形可能太大，其像会跨过目标中的多个单形，无法由顶点像定义一个单纯映射。细分使顶点开星变小，连续性保证每个小开星的像可落进目标某个顶点的开星。

English. An original simplex may be too large: its image can cross several target simplices, so its vertex images may not define a simplicial map. Subdivision makes the vertex stars smaller, and continuity ensures that each small star can map into an open star of a target vertex.

4 第 4 章：基本群 | Chapter 4: The fundamental group

4.1 路径、闭路同伦与群运算 | Paths, loop homotopy, and the group operation

路径是连续映射 $\alpha : [0,1] \to X$。若 $\alpha(0) = \alpha(1) = x_0$，称它为基于 x_0 的闭路。若两条端点相同的路径可以在始终固定端点的连续变形中互相变到，称它们端点固定同伦。基本群 $\pi_1(X, x_0)$ 是基于 x_0 的闭路按端点固定同伦取等价类后得到的集合，运算是路径拼接。

English. A path is a continuous map $\alpha : [0,1] \to X$. It is a loop based at x_0 if $\alpha(0) = \alpha(1) = x_0$. Two paths with the same endpoints are endpoint-fixed homotopic if one can be continuously deformed into the other while keeping both endpoints fixed. The fundamental group $\pi_1(X, x_0)$ consists of based loops modulo endpoint-fixed homotopy, with path concatenation as its operation.

若 $\alpha(1) = \beta(0)$，先走完 α 再走 β 的拼接定义为

公式名称 / Formula name. 路径拼接公式 / the path concatenation formula

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t), & 0 \le t \le \frac{1}{2}, \\ \beta(2t-1), & \frac{1}{2} \le t \le 1. \end{cases}$$

公式英文读法 / How to read it. “Alpha star beta at t equals alpha of two t for t from zero to one half, and beta of two t minus one for t from one half to one.”

English. When $\alpha(1) = \beta(0)$, the concatenation traverses α first and then β, using half of the parameter interval for each path.

常值闭路是单位元，反向路径 $\bar{\alpha}(t) = \alpha(1-t)$ 给出逆元。拼接在路径本身层面依赖参数化，但结合律在同伦类层面成立，因此等价类构成群。基本群可能非交换，因为先走 α 再走 β，一般不能连续变成先走 β 再走 α。

English. The constant loop is the identity, and the reversed path $\bar{\alpha}(t) = \alpha(1-t)$ gives an inverse. Concatenation depends on parametrization at the path level, but it is associative up to homotopy, so the equivalence classes form a group. The fundamental group need not be commutative: traversing α and then β may not be homotopic to traversing them in the opposite order.

4.2 圆周的基本群与绕数 | The fundamental group of the circle and winding number

对基点为 1 的圆周闭路 $\gamma : [0,1] \to S^1$，可以沿参数连续选取辐角 $\theta(t)$，使 $\gamma(t) = e^{i\theta(t)}$。由于起点和终点都映到 1，辐角变化量是 2π 的整数倍。这个整数是闭路的绕数：

公式名称 / Formula name. 绕数公式 / the winding number formula

$$\mathrm{wind}(\gamma) = \frac{\theta(1) - \theta(0)}{2\pi} \in \mathbb{Z}.$$

公式英文读法 / How to read it. “The winding number of gamma is theta at one minus theta at zero, divided by two pi.”

English. For a loop $\gamma : [0,1] \to S^1$ based at 1, choose a continuous argument $\theta(t)$ such that $\gamma(t) = e^{i\theta(t)}$. Since the endpoints of the loop are both 1, the change in argument is an integer multiple of 2π. That integer is its winding number, given by the displayed formula.

绕数在端点固定同伦下不变，拼接路径时相加。绕数为零的圆周闭路可以缩成常值路；对每个整数 n，闭路 $\gamma_n(t) = e^{2\pi i n t}$ 的绕数为 n。因此绕数给出群同构 $\pi_1(S^1, 1) \cong \mathbb{Z}$。相反，可缩空间（如圆盘或 $\mathbb{R}^m$）的所有闭路都能缩成常值路，其基本群平凡。

English. Winding number is invariant under endpoint-fixed homotopy and adds under concatenation. A loop with winding number zero contracts to the constant loop. For every integer n, the loop $\gamma_n(t) = e^{2\pi i n t}$ has winding number n. Thus winding number gives an isomorphism $\pi_1(S^1, 1) \cong \mathbb{Z}$. In a contractible space such as a disk or $\mathbb{R}^m$, every loop contracts, so the fundamental group is trivial.

例子 / Example. 沿圆周逆时针走两圈的闭路绕数为 2，顺时针走一圈的绕数为 -1。先走前者再走后者，拼接后的绕数为 $2 + (-1) = 1$。因此这两条路径的拼接类对应整数 1，而不是“走过的总路程”。

English. A loop going twice counterclockwise has winding number 2, while a clockwise loop has winding number -1. Concatenating them gives winding number $2 + (-1) = 1$. The integer records net winding, not the total distance travelled.

可能问题 / Possible question. 为什么基本群可以证明圆周与圆盘不同胚？

English. How can the fundamental group show that the circle and disk are not homeomorphic?

回答要点 / Key answer. 圆周的基本群是 $\mathbb{Z}$，圆盘可缩所以基本群平凡。同胚会诱导基本群同构，而这两个群不同构，因此圆周与圆盘不同胚。

English. The circle has fundamental group $\mathbb{Z}$, while the disk is contractible and has trivial fundamental group. A homeomorphism induces an isomorphism of fundamental groups, and these groups are not isomorphic. Hence the circle and disk are not homeomorphic.

4.3 Seifert–van Kampen 定理与生成元、关系 | The Seifert–van Kampen theorem and presentations

基本群的直接闭路计算有时很困难。**Seifert–van Kampen 定理**的作用是把空间拆成较简单的重叠部分再计算。若 $X = U \cup V$，$U, V, U \cap V$ 路径连通，且基点 $x_0 \in U \cap V$，则 $\pi_1(X, x_0)$ 由 $\pi_1(U, x_0)$ 与 $\pi_1(V, x_0)$ 的自由积生成，并把交集中的同一闭路在两边的像识别起来。换言之，若 i_U, i_V 是包含映射，需商去由 $i_{U*}(g)i_{V*}(g)^{-1}$ 生成的正规子群，其中 $g \in \pi_1(U \cap V, x_0)$。

English. The Seifert–van Kampen theorem computes a fundamental group by decomposing a space into simpler overlapping pieces. If $X = U \cup V$, the sets $U, V, U \cap V$ are path-connected, and the basepoint lies in the intersection, then $\pi_1(X, x_0)$ is generated by the free product of $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$, with loops from the intersection identified through the two inclusion maps. Equivalently, one quotients by the normal subgroup generated by $i_{U*}(g)i_{V*}(g)^{-1}$ for loops g in the intersection.

这一定理常把基本群写成**生成元与关系**。例如，环面可表示为一个顶点、两条边 a, b 和一个二维面；一维骨架是两个圆周一点粘合，基本群为自由群 $F(a, b)$。二维面的边界按粘合顺序是 $aba^{-1}b^{-1}$，加入该面相当于令这个边界词为单位元。因此

公式名称 / Formula name. 环面基本群的表示 / the presentation of the fundamental group of the torus

$$\pi_1(T^2) = \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

公式英文读法 / How to read it. “The fundamental group of the torus is generated by a and b, subject to the relation a b a inverse b inverse equals the identity.”

English. The torus has a cell structure with one vertex, two edges a, b, and one two-dimensional face. Its one-dimensional skeleton is a wedge of two circles, with free fundamental group $F(a, b)$. The boundary of the face is the word $aba^{-1}b^{-1}$. Attaching the face makes this word trivial, giving the displayed presentation. The relation says that a and b commute, so the group is $\mathbb{Z}^2$.

应用 van Kampen 时，不能只看图就猜关系式；必须给出满足定理条件的开集分解或等价的细胞粘合论证。基本群不同构足以说明空间不同伦等价，因而不同胚；基本群同构则通常不能反推出空间同胚。

English. To use van Kampen, a diagram alone is not enough: one must provide an open-set decomposition satisfying the theorem or an equivalent cell-attachment argument. Nonisomorphic fundamental groups show that spaces are not homotopy equivalent and hence not homeomorphic. Isomorphic fundamental groups usually do not imply that spaces are homeomorphic.

4.4 基本群与第一同调 | The fundamental group and first homology

对路径连通的空间，第一同调群是基本群的交换化：

公式名称 / Formula name. 第一同调与基本群交换化定理 / the theorem identifying first homology with the abelianization of the fundamental group

$$H_1(X; \mathbb{Z}) \cong \pi_1(X, x_0)^{\mathrm{ab}} = \pi_1(X, x_0)/[\pi_1(X, x_0), \pi_1(X, x_0)].$$

公式英文读法 / How to read it. “The first homology group is the fundamental group modulo its commutator subgroup.”

English. For a path-connected space, the first homology group is the abelianization of the fundamental group: quotient by the commutator subgroup generated by elements $aba^{-1}b^{-1}$.

直观上，基本群保留闭路拼接的先后顺序，可能非交换；同调把闭路转成可线性相加的链，因此交换化并丢失一部分信息。两个圆周一点粘合成的花束有 $\pi_1 = F(a,b)$，其中 ab 与 ba 不同；但 $H_1 \cong \mathbb{Z}^2$。环面也有 $H_1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$，所以第一同调相同并不意味着基本群相同。

English. The fundamental group retains the order of loop concatenation and may be noncommutative. Homology turns loops into chains that can be added, so it abelianizes the group and loses some information. The wedge of two circles has $\pi_1 = F(a,b)$, where ab and ba differ, but $H_1 \cong \mathbb{Z}^2$. The torus also has $H_1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$, so equal first homology does not imply equal fundamental groups.

可能问题 / Possible question. 基本群与第一同调的主要区别是什么？

English. What is the main difference between the fundamental group and first homology?

回答要点 / Key answer. 基本群记录基点闭路的同伦类及其拼接，可能非交换；第一同调是基本群的交换化，只保留可交换的环信息。因而 H_1 通常较容易计算，但信息比 π_1 少。

English. The fundamental group records based-loop homotopy classes and their concatenation, which may be noncommutative. First homology is its abelianization and retains only the commutative loop information. Thus H_1 is often easier to compute but contains less information than π_1.

5 第 5 章：覆盖空间 | Chapter 5: Covering spaces

5.1 覆盖映射与均匀覆盖邻域 | Covering maps and evenly covered neighborhoods

连续满射 $p : \widetilde{X} \to X$ 称为覆盖映射，如果每个 $x \in X$ 都有开邻域 U，使得原像 $p^{-1}(U)$ 是互不相交开集 V_α 的并，且每个限制 $p|_{V_\alpha} : V_\alpha \to U$ 都是同胚。U 称为均匀覆盖邻域；$p^{-1}(x)$ 称为 x 上方的纤维。覆盖空间在每个足够小的邻域上看起来像若干份底空间，但全局的粘合方式可以不同。

English. A continuous surjection $p : \widetilde{X} \to X$ is a covering map if every $x \in X$ has an open neighborhood U such that $p^{-1}(U)$ is a disjoint union of open sets V_α, each mapped homeomorphically onto U by the restriction of p. Such a set U is evenly covered, and $p^{-1}(x)$ is the fiber over x. Locally a covering looks like several copies of the base, although the global arrangement can differ.

例子 / Example. 映射 $p : \mathbb{R} \to S^1$，$p(t) = e^{2\pi i t}$，是圆周的覆盖。取圆周上一段不绕满一圈的小弧，它的原像是实线上相隔整数距离的一列互不相交区间；每个区间都通过 p 同胚映到这段小弧。每个圆周点上方有无穷多个点。

English. The map $p : \mathbb{R} \to S^1$, $p(t) = e^{2\pi i t}$, is a covering of the circle. A short arc that does not make a full turn has as its preimage a disjoint sequence of intervals in $\mathbb{R}$, separated by integer translations. Each interval maps homeomorphically onto the arc, and every point of the circle has infinitely many preimages.

5.2 路径提升与同伦提升 | Path lifting and homotopy lifting

路径提升性质。给定覆盖 $p : \widetilde{X} \to X$、路径 $\gamma : [0,1] \to X$ 和起点 $\widetilde{x}_0 \in p^{-1}(\gamma(0))$，存在唯一连续路径 $\widetilde{\gamma} : [0,1] \to \widetilde{X}$ 满足 $p \circ \widetilde{\gamma} = \gamma$ 且 $\widetilde{\gamma}(0) = \widetilde{x}_0$。直观上，将区间 $[0,1]$ 分成若干小段，使每段路径落在某个均匀覆盖邻域中；从已选定的起始片开始，每一小段都只能继续到唯一相接的覆盖片。

English. The path lifting property states that for a covering $p : \widetilde{X} \to X$, a path $\gamma : [0,1] \to X$, and a chosen starting point $\widetilde{x}_0$ above $\gamma(0)$, there is a unique path $\widetilde{\gamma}$ with $p \circ \widetilde{\gamma} = \gamma$ and $\widetilde{\gamma}(0) = \widetilde{x}_0$. To see why, subdivide the interval so that each piece of the path lies in an evenly covered neighborhood. Once the starting sheet is chosen, each successive piece has a unique compatible lift.

同伦提升性质。若底空间中的路径随参数连续变形，并已给定初始路径的提升，则整个变形也能连续提升。特别地，端点固定同伦的两条闭路从同一覆盖点出发时，提升终点相同。因此，提升闭路后是否回到起点，以及落到纤维中的哪个点，能检测闭路的同伦信息。

English. The homotopy lifting property says that a continuous deformation of paths lifts continuously once a lift of the initial path is fixed. In particular, endpoint-fixed homotopic loops starting from the same point of the cover have lifts with the same endpoint. Whether the lifted loop returns to its starting point, and which point of the fiber it reaches, detects information about the loop's homotopy class.

例子 / Example. 将圆周上正向绕一圈的闭路从实线上的 0 提升，得到 $\widetilde{\gamma}(s) = s$，终点是 1，没有回到起点；绕 n 圈时终点是 n。因此在 $\mathbb{R} \to S^1$ 这个覆盖下，提升终点正好记录绕数，且提升闭合当且仅当绕数为零。

English. Lift a positively oriented loop around the circle from $0 \in \mathbb{R}$. Its lift is $\widetilde{\gamma}(s) = s$, ending at 1 rather than returning to its starting point. A loop making n turns ends at n. Thus the endpoint of the lift under $\mathbb{R} \to S^1$ records the winding number, and the lift is closed exactly when the winding number is zero.

5.3 覆盖与基本群子群的对应 | The correspondence between covers and subgroups

设底空间 X 连通、局部路径连通且半局部单连通。半局部单连通的意思是：每一点都有足够小的邻域，使其中的闭路在整个 X 中可缩；它不是说这个邻域内的闭路必须在邻域自身中可缩。此条件保证标准的覆盖分类定理适用。固定基点 x_0 后，带基点的连通覆盖（按保持基点的覆盖同构分类）与 $\pi_1(X, x_0)$ 的子群 H 一一对应；不指定覆盖基点时，对应于子群的共轭类。

English. Assume that the base space X is connected, locally path-connected, and semilocally simply connected. Semilocal simple connectivity means that each point has a sufficiently small neighborhood whose loops contract in the whole space; the loops need not contract inside that neighborhood itself. These hypotheses ensure that the standard classification theorem applies. With a basepoint x_0, based connected covers, up to based isomorphism, correspond to subgroups H of $\pi_1(X, x_0)$. Without a chosen point in the cover, they correspond to conjugacy classes of subgroups.

对覆盖 $p : (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$，对应子群可写成

公式名称 / Formula name. 覆盖对应子群的刻画 / the subgroup associated with a based covering

$$H = p_*\big(\pi_1(\widetilde{X}, \widetilde{x}_0)\big) \leq \pi_1(X, x_0).$$

公式英文读法 / How to read it. "H is the image of the fundamental group of the covering space under the homomorphism induced by p."

English. The subgroup associated with a based cover is the image of its fundamental group under the homomorphism induced by the covering map. Equivalently, it consists of loop classes in the base whose lifts from $\widetilde{x}_0$ are closed.

覆盖的片数等于子群指数 $[\pi_1(X, x_0) : H]$。$H = \{1\}$ 对应的连通覆盖是普遍覆盖：它本身单连通。这个对应把“底空间中哪些闭路提升后闭合”转成基本群中的子群问题。

English. The number of sheets of the cover equals the subgroup index $[\pi_1(X, x_0) : H]$. The connected cover corresponding to the trivial subgroup is the universal cover, which is simply connected. This correspondence turns the question of which loops lift to closed loops into a subgroup problem in the fundamental group.

5.4 圆周覆盖的完整例子 | A complete example: coverings of the circle

圆周的基本群是 $\mathbb{Z}$，其子群只有 $n\mathbb{Z}$（$n \geq 1$）和 $\{0\}$。对 $n \geq 1$，映射 $p_n : S^1 \to S^1$ 定义如下：

公式名称 / Formula name. 圆周的 n 层覆盖 / the n-sheeted covering map of the circle

$$p_n(z) = z^n.$$

公式英文读法 / How to read it. “P sub n maps z to z to the n-th power.”

English. For each positive integer n, the map $p_n(z) = z^n$ is an n-sheeted covering of the circle.

每个底空间点恰有 n 个原像，所以这是 n 层覆盖；它在基本群上诱导 $k \mapsto nk$，像子群为 $n\mathbb{Z}$，指数为 n。提升一圈的闭路不会闭合，必须绕底圆 n 圈才回到原覆盖点。子群 $\{0\}$ 的指数无穷，对应普遍覆盖 $\mathbb{R} \to S^1$。这给出圆周全部连通覆盖的分类：有限层的圆周覆盖和实线普遍覆盖。

English. Every point in the base circle has exactly n preimages, so this is an n-sheeted cover. The induced map on fundamental groups is $k \mapsto nk$, whose image is $n\mathbb{Z}$ and whose index is n. A lift of one turn is not closed; the lift closes after the base loop makes n turns. The trivial subgroup $\{0\}$ has infinite index and corresponds to the universal cover $\mathbb{R} \to S^1$. These are all connected covers of the circle: finite-sheeted circle covers and the universal cover by the real line.

可能问题 / Possible question. 如何用覆盖空间读出圆周闭路的绕数？

English. How can a covering space reveal the winding number of a loop in the circle?

回答要点 / Key answer. 把闭路从实线覆盖中的指定点开始提升。若底圆绕 k 圈，提升终点相对起点平移整数 k；因此终点位移就是绕数，提升闭合当且仅当 $k = 0$。

English. Lift the loop from a chosen point in the real-line cover. If it winds k times around the base circle, its lift ends at the starting point translated by the integer k. This displacement is the winding number, and the lift closes exactly when $k = 0$.